

Part 1: Exploring Exponential Distributions and Their Center and Spread

Kimberly Bormann

2025-04-05

Overview

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Distributions

We will create two simulations to investigate and compare their shape, mean, and variance.

First Distribution - Single Sample of Exponential Variables

In this report we will explore two distributions. The first is a 1000 random exponential variables with $\lambda = 0.2$. Given our knowledge of exponential functions, we expect that when we visualize this distribution it will look very similar to the function used to generate the random variables which is $f(x) = 0.2e^{-0.2x}$, where x is randomly generated.

First we generate the variables.

```
#set lambda value
lambda <- 0.2
#simulation of 1000 random exponential values
exp_raw <- rexp(1000, rate = lambda)
#report
cat("First few entries of a single simulation of 1000 values:\n")
```

First few entries of a single simulation of 1000 values:

```
head(exp_raw)
```

```
## [1] 4.2172863 2.8830514 6.6452743 0.1578868 0.2810549 1.5825061
```

Now, let's do the simple calculations for the theoretical means and variance for an exponential distribution with 1000 occurrences and $\lambda = 0.2$.

```
theoretical_mean <- 1/lambda
cat("The theoretical mean is", theoretical_mean, "\n")
```

The theoretical mean is 5

```
theoretical_sd <- 1/lambda
theoretical_var <- (1/lambda)^2
cat("The theoretical variance is", theoretical_var, "\n")
```

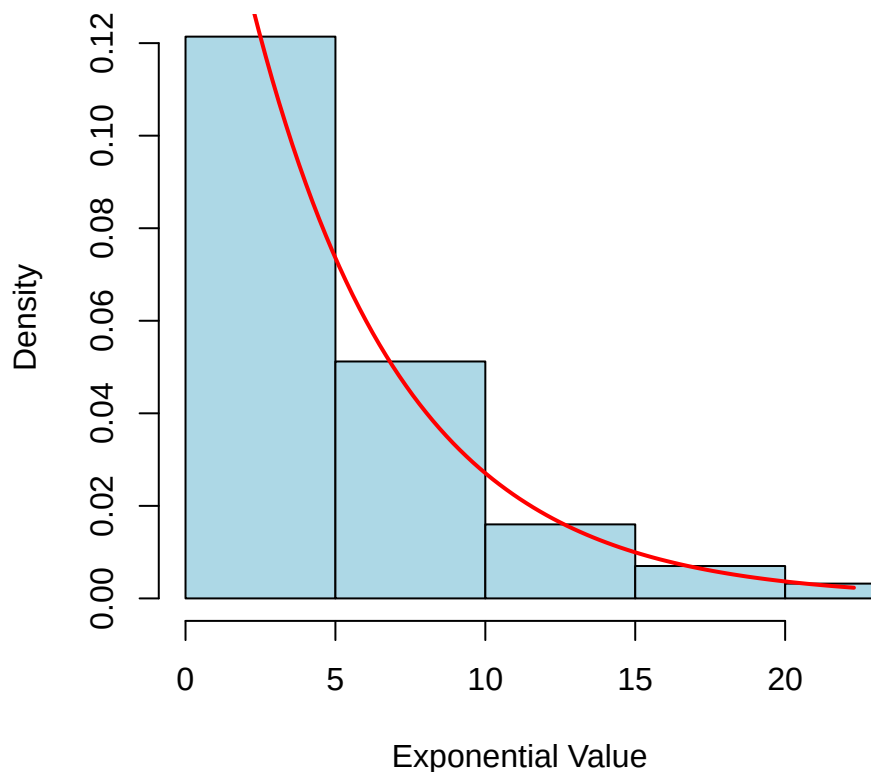
The theoretical variance is 25

Then we can view the distribution to see if it reflects our expectation that it is right skewed like the decreasing exponential function. To make this easier, I overlay the function $f(x) = 0.2e^{-0.2x}$ on the histogram.

```
#histogram form 1000 exponential values
hist(exp_raw, probability = TRUE,
     main = "Histogram of 1000 Exponential Values",
     xlab = "Exponential Value",
     col = "lightblue",
     xlim = c(0, quantile(exp_raw, 0.99)))

#overlay the theoretical exponential density curve
curve(dexp(x, rate = lambda),
     from = 0,
     to = quantile(exp_raw, 0.99),
     add = TRUE,
     col = "red",
     lwd = 2)
```

Histogram of 1000 Exponential Values



This density function is right-skewed. We can see that the random sample is a very close match to the theoretical density function because the Law of Large Numbers (LLN) tells us that the larger the sample the more likely it is to approach the theoretical density curve.

Lastly, let's compare the theoretical mean and variance to the sample mean and variance.

Mean of this single simulation: 5.1498964 Variance of a single simulation: 25.2207028

We can see that these the sample and theoretical means and variance are very similar.

#Second Distribution - 1000 Sample of 40 Exponential Variables

The theoretical means and variance for 1000 samples of 40 exponential random variables with $\lambda = 0.2$.

```
#theoretical mean calculation
lambda <- 0.2
theoretical_mean <- 1/lambda
cat("The theoretical mean is", theoretical_mean, "\n")
```

```
## The theoretical mean is 5
```

```
theoretical_sd <- 1/lambda
```

```
#theoretical variance calculation
n <- 40
theoretical_sd <- 1/lambda
theoretical_se <- theoretical_sd/sqrt(n)
theoretical_mean_var <- theoretical_se^2
cat("The theoretical variance of the means is", theoretical_mean_var, "\n")
```

```
## The theoretical variance of the means is 0.625
```

Next, let's generate the values and compare the mean and variance of the sample to the theoretical mean and variance.

```
#generate 1000 simulation of 40 random exponential variables in a matrix where each column contains 40
n_sim <- 1000
sim_matrix <- matrix(rexp(n*n_sim, rate = lambda), nrow = n, ncol = n_sim)
#calculate the mean of each sample
sim_means <- apply(sim_matrix, 2, mean)
#calculate the variance of each sample
sim_vars <- apply(sim_matrix, 2, var)
#vector of all the sample means
avg_values <- replicate(1000, mean(rexp(n, rate = lambda)))
cat("first few sample means:\n")
```

```
## first few sample means:
```

```
head(sim_means)
```

```
## [1] 4.801996 4.426148 4.417217 4.967223 5.470471 5.657978
```

Now let's compare the sample means and variance to the theoretical ones calculated earlier.

```
cat("Sample Means (averages of 40 exponentials):\n")
```

```
## Sample Means (averages of 40 exponentials):
```

```
cat(" Sample Mean:", round(mean(sim_means), 3), "\n")
```

```
## Sample Mean: 5.013
```

```
cat(" Sample Variance:", round(var(sim_means), 3), "\n\n")
```

```
## Sample Variance: 0.604
```

```
cat("Theoretical Values:\n")
```

```
## Theoretical Values:
```

```
cat(" Theoretical Mean:", theoretical_mean, "\n")
```

```
## Theoretical Mean: 5
```

```
cat(" Theoretical Variance of the Sample Mean:", round(theoretical_mean_var, 3), "\n")
```

```
## Theoretical Variance of the Sample Mean: 0.625
```

As you can see the sample mean and variance is extremely close to the theoretical mean and variance.

Lastly, let's view the distribution. Unlike the first simulation which generated 1000 random exponential variables, this simulation generated 1000 means of random exponential variables.

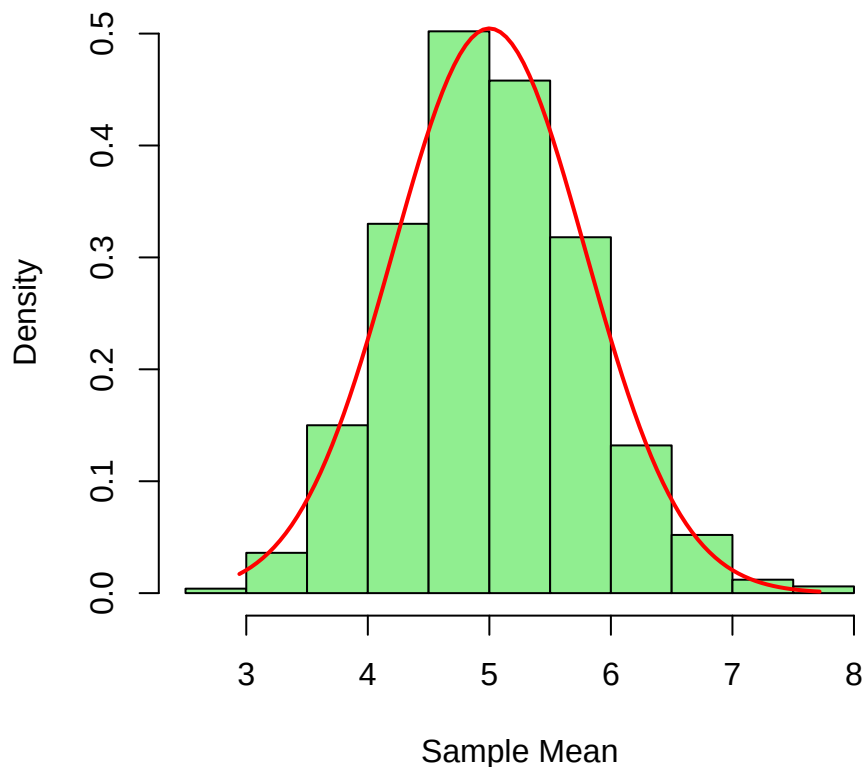
Therefore, by the Central Limit Theorem (CLT), we expect that this distribution will be normally distributed because CLT states that, although each individual observation is not normally distributed, the distribution of the sample mean will tend toward a normal distribution as the sample size increases. With sample size 1000 we expect a mostly normal distribution.

For comparison, I overlay the normal distribution with the theoretical mean and variance to the sample distribution.

```
#plot the histogram of sample means with density scaling
hist(sim_means, probability = TRUE,
     main = "Histogram of 1000 Sample Means (n = 40)",
     xlab = "Sample Mean",
     col = "lightgreen")

#overlay the theoretical normal density curve
curve(dnorm(x, mean = theoretical_mean, sd = sqrt(theoretical_mean_var)),
     from = min(sim_means), to = max(sim_means),
     add = TRUE, col = "red", lwd = 2)
```

Histogram of 1000 Sample Means (n = 40)



We can see that the distribution is normally distributed as expected.